

CS212 Review Problems: GUI, Events, and Exceptions (Lectures 16–18)

CS212

Instructions

Answer the following questions. Show reasoning where appropriate.

Multiple Choice Questions

1. What best describes a GUI program?
 - A. It executes statements strictly from top to bottom
 - B. It repeatedly loops through all code continuously
 - C. It waits for events and responds when they occur
 - D. It only runs when the user presses a button
2. What is a callback?
 - A. A method that is called immediately when defined
 - B. A function passed to be executed later when an event occurs
 - C. A function that returns another function
 - D. A method that runs automatically at program start
3. Why do we use a `JPanel` inside a `BorderLayout`?
 - A. To improve performance
 - B. Because `BorderLayout` cannot display buttons
 - C. To allow multiple components in one region
 - D. To automatically resize all components
4. What does `text.trim()` do?
 - A. Removes all spaces
 - B. Removes leading and trailing whitespace
 - C. Splits the string
 - D. Converts to lowercase

5. What does `text.split("\\s+")` do?
 - A. Splits into characters
 - B. Removes whitespace
 - C. Splits at commas
 - D. Splits into words using whitespace
6. When an exception is thrown, what happens?
 - A. Program always crashes
 - B. Method continues normally
 - C. Execution pauses
 - D. Execution stops and jumps to a catch block
7. Where should computations be performed?
 - A. GUI class
 - B. Model class
 - C. Event listener
 - D. Main method
8. What is the role of the GUI?
 - A. Perform computations
 - B. Store data
 - C. Handle interaction and display
 - D. Manage files
9. Why separate file IO into its own class?
 - A. Avoid exceptions
 - B. Reduce memory
 - C. Simplify layout
 - D. Separate responsibilities
10. What does a `throws` clause do?
 - A. Handles exception
 - B. Prevents exception
 - C. Declares it may pass to caller
 - D. Logs error
11. What does `frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE)` do?
 - A. Closes all running programs

- B. Hides the window only
 - C. Defines behavior when the window is closed
 - D. Restarts the application
12. What is the purpose of `addActionListener`?
- A. To draw components
 - B. To attach behavior to an event
 - C. To create a button
 - D. To resize components
13. What does `new JScrollPane(area)` do?
- A. Creates a new window
 - B. Wraps the text area to allow scrolling
 - C. Changes font size
 - D. Adds buttons
14. What is the default layout of a `JPanel`?
- A. `BorderLayout`
 - B. `GridLayout`
 - C. `FlowLayout`
 - D. Null layout
15. What does `setPreferredSize()` do?
- A. Forces exact size
 - B. Suggests a size to the layout manager
 - C. Resizes the frame
 - D. Removes layout control
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Short Answer

- a. What does it mean for a program to be event-driven?
 - b. When does a callback run?
 - c. What is a functional interface?
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Programming Problem 1 (Simple GUI)

Write a GUI with:

- A button labeled ‘Click Me’
- A label initially showing ‘Ready’

When the button is clicked, the label should change to “Clicked”.

```
// Your code here
```

—

Programming Problem 2 (Simple GUI)

Write a GUI with:

- A JTextField
- A button labeled “Show Length”
- A label

When the button is clicked, display the length of the text in the label.

```
// Your code here
```

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Programming Problem 3 (GUI + Model)

Design a small application:

Features:

- A JTextArea for input
- A button “Longest Word”
- A label for output

Behavior:

- When the button is clicked, find the longest word in the text
- If no words exist, throw an exception

Tasks:

1. Write a model class with a method:

```
String longestWord(String text) throws Exception
```

2. Write the GUI handler that calls the model and handles exceptions